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The EARTH Energy Efficiency Evaluation Framework (E^3F):

A methodology to evaluate radio network energy efficiency at system level

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Magnus Olsson, Ericsson AB, Stockholm, Sweden
Technical Manager, EARTH project
magnus.a.olsson@ericsson.com

SEVENTH FRAMEWORK PROGRAMME

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Acknowledgements

This presentation is based on material from the following paper:

"How much energy is needed to run a wireless network?"

by G. Auer (DOCOMO Eurolabs), V. Giannini and C. Desset (IMEC),

I. Godor, P.Skillermark and M. Olsson (Ericsson), M. Imran (University of Surrey), D. Sabella (Telecom Italia), M. Gonzalez (TTI), O. Blume (Alcatel-Lucent), and A. Fehske (TU Dresden)

published in ***IEEE Wireless Communications Magazine, October 2011***

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Background and motivation

earth
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- **What is EARTH?**
  - EU-funded project with 15 partners from industry and academia
  - Started in January 2010, will end in June 2012
  - Jointly led by Alcatel-Lucent (PC) and Ericsson (TM)
- **EARTH has an ambitious overall goal to decrease the radio access network (RAN) energy consumption by 50%**
  - More efficient hardware components, RRM mechanisms, network management solutions, deployment strategies, etc.
- **Evaluation methodology for energy consumption**
  - The EARTH **E**nergy **E**fficiency **E**valuation **F**ramework (E^3F)
  - Primarily focused on a given system, with and without feature(s)
  - Captures network dynamics and behaviour
  - Fully compatible with existing 3GPP and ITU-R evaluation methodology

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EARTH Energy Efficiency Evaluation Framework (E^3F)

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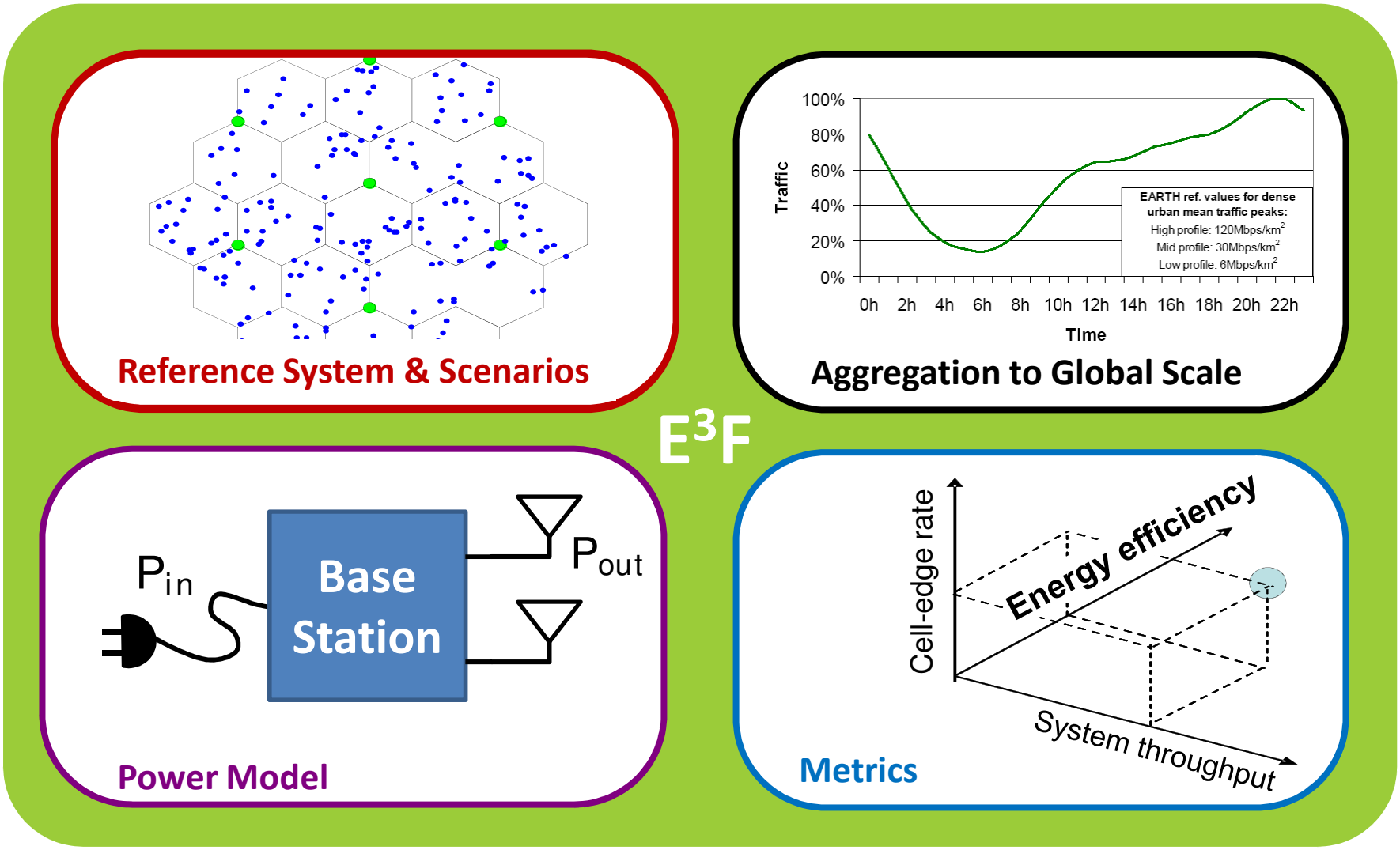

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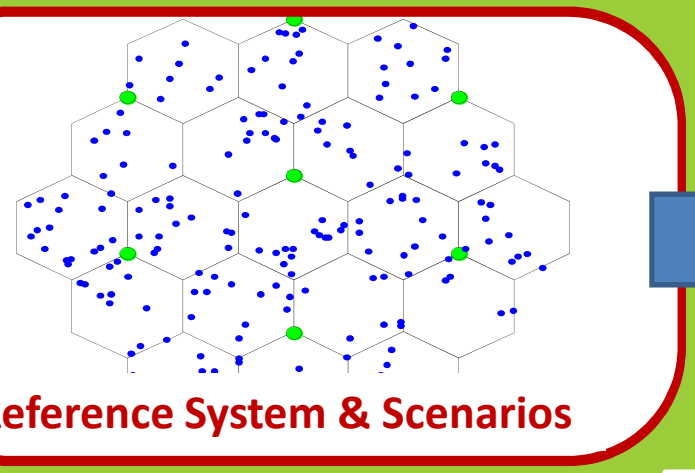
Reference system and scenarios



Basis for traditional (small-scale, short-term) **System Level Simulations**

E³F

- Define system parameters and scenarios so to ensure comparable results
- Dynamic system simulation for different deployments
  - Dense urban, urban, suburban and rural
- 3GPP and ITU-R compliant evaluation methods

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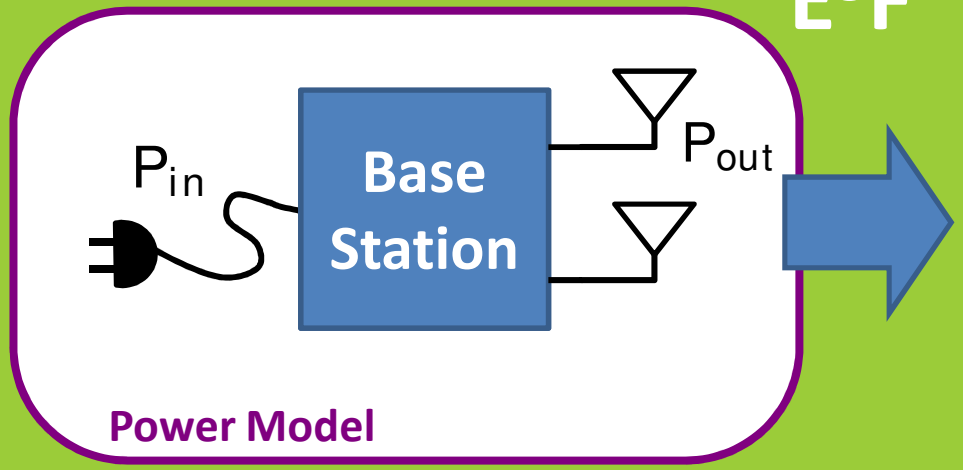
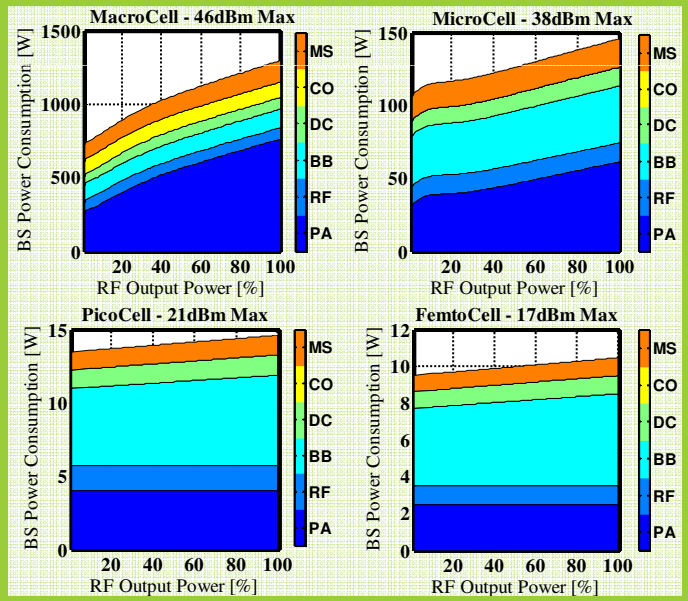
Base station (BS) power model

- Maps RF output power P_{out} to the total BS supply power P_{in}
- Power breakdown of different components for various BS types (macro, micro, pico, femto BS)
  - → Includes power amplifier (PA), RF, base band (BB) processing, DC-DC unit, cooling, mains supply
- Power model contributed to 3GPP RAN1

Realistic **Power Model**

E³F





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EARTH energy efficiency metrics

Recommended use of **Energy Consumption Metrics**

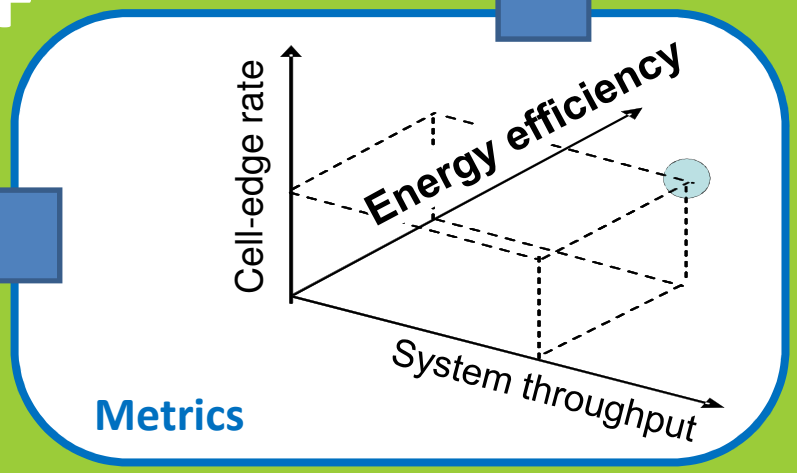
- [J/bit] – reflecting traditional capacity and data rate improvements
  - Increasing peak data rates quickly decreases [J/bit]
- [W/m^2] – reflecting crucial coverage and service area aspects
  - High fixed cost for coverage makes [W/m^2] very important
- Metrics communicated to ETSI TC EE

Performance Metrics
for Concept Evaluations

- Performance metrics follow 3GPP recommendations [TR36.814]
- Served cell throughput
- Mean, 5th, 10th, and 50th percentile user throughput

  → Cell-edge throughput

E^{3}F



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Aggregation to global scale (1/2)

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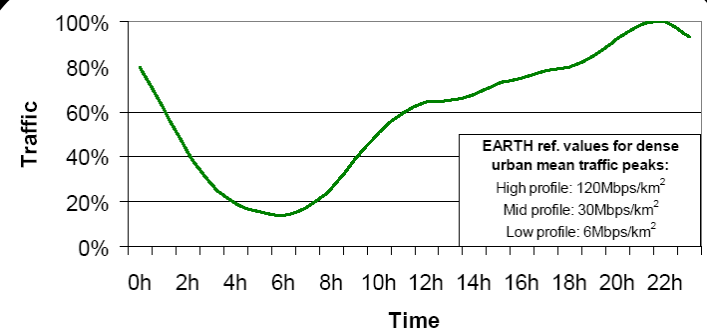
Country wide 24 hour **Traffic & Deployment Model**



Aggregation to Global Scale

E³F

- The EARTH daily traffic profile is extracted from operator data, and represents an exemplary European country
- Based on network traffic data; geographical and population data

➔ Models average traffic aggregation to 24 hours

➔ Maps reference scenarios to country and European scale

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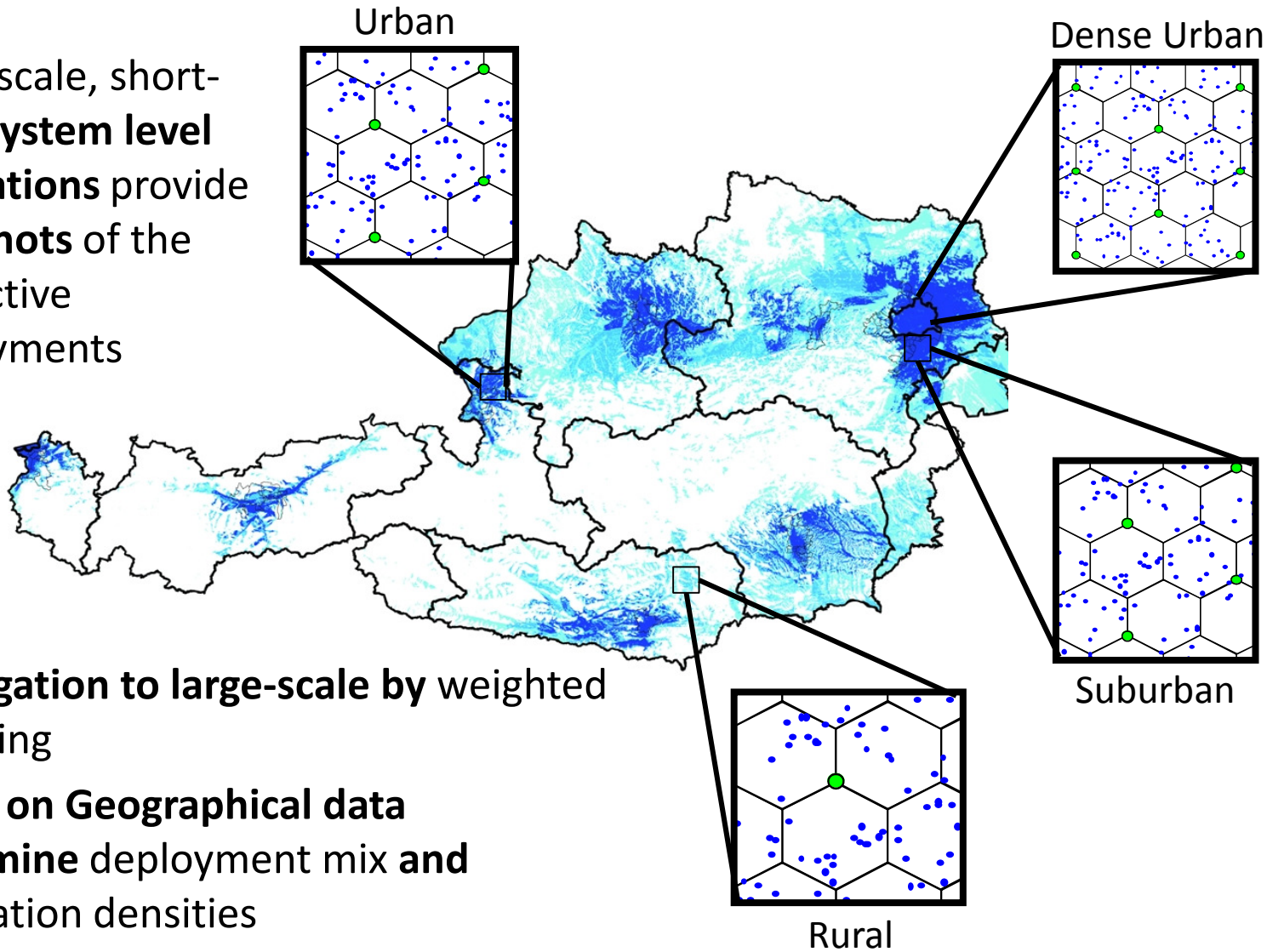
Aggregation to global scale (2/2)

- Small-scale, short-term **system level simulations** provide **snapshots** of the respective deployments



- **Aggregation to large-scale by** weighted summing

➔ **Based on Geographical data determine** deployment mix **and** population densities

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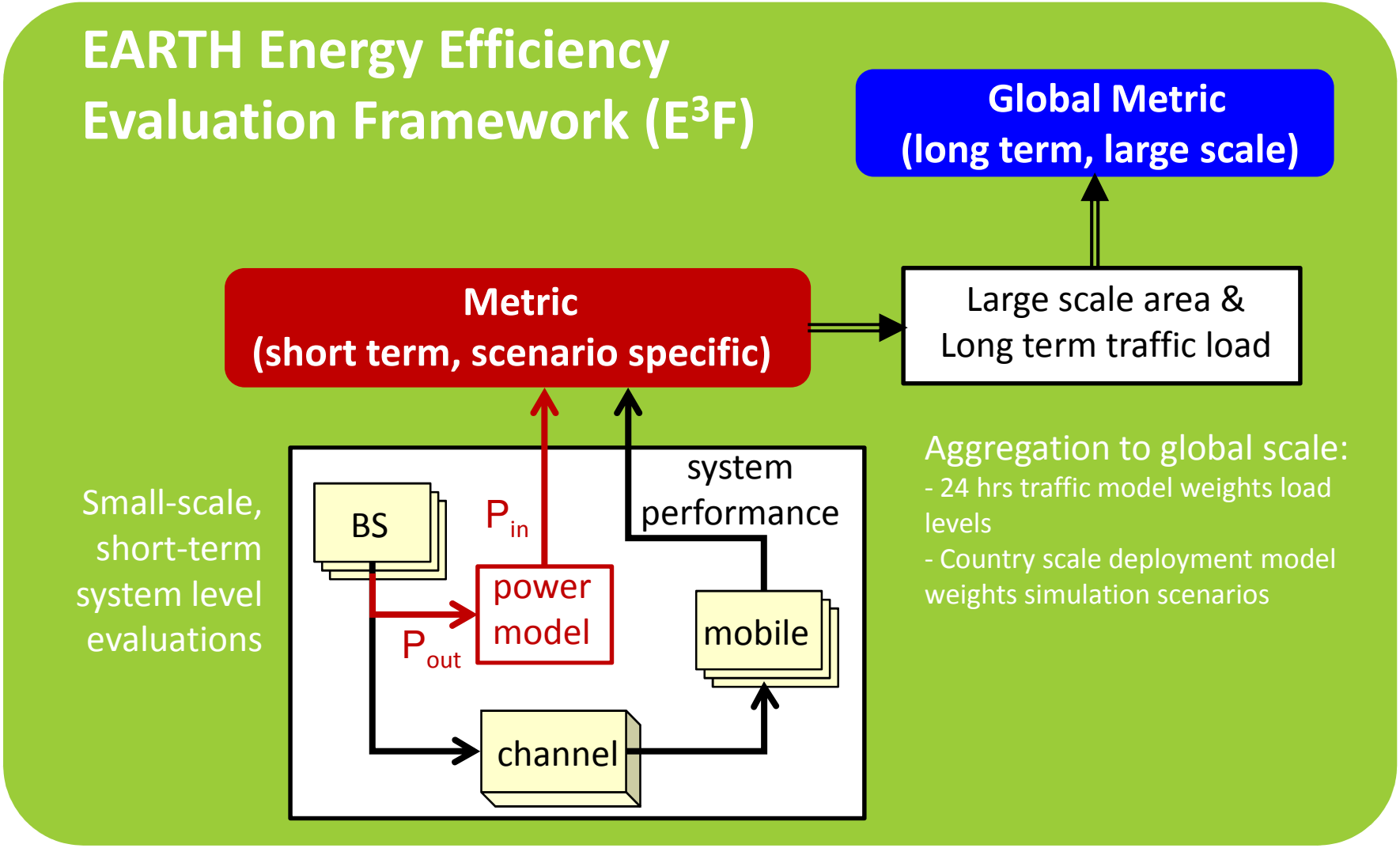
E^3F – Summarizing flow chart



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Case study: Energy efficiency of LTE

Objective:

- Assess the energy efficiency of a typical LTE network, for a typical operator, in a typical European country

Purpose:

- Situation analysis, identify improvement areas

Short-term, small-scale evaluations:

- Macro-cellular network with regular hexagonal cell layout is implemented
  → 19 sites, each with 3 sectors
- Inter-site distance (ISD)
  - Dense urban and urban environments: 500 m
  - Suburban and rural areas: 1732 m
- The users are uniformly distributed, with user densities corresponding to the respective deployment scenarios
- The simulation parameters are taken from 3GPP reports [TR36.814], [TR25.814]
  - 10 MHz bandwidth
  - 2.1 GHz carrier frequency
  - 2x2 MIMO transmission with adaptive rank adaptation

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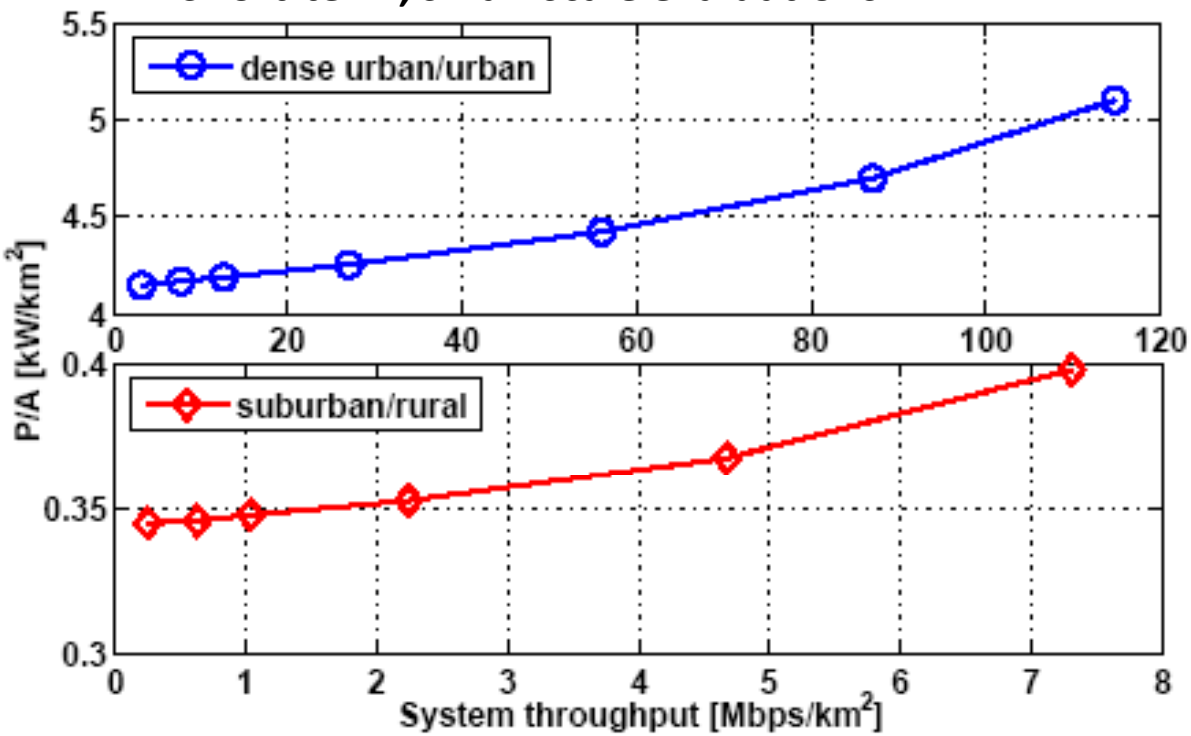
Power Consumption per Unit Area (P/A)

Short-term, small-scale evaluations



• Only a modest dependency of the system throughput on the network power consumption

➔ Due to macro BS power model, which exhibits **poor efficiency at low loads**

- **Urban scenario** achieves more than 10 times higher system throughput than **suburban/rural scenarios** → due to higher site density (urban ISD=500m, compared to suburban/rural ISD=1732m)
- However, **suburban/rural scenarios** consume more than 10 times less power

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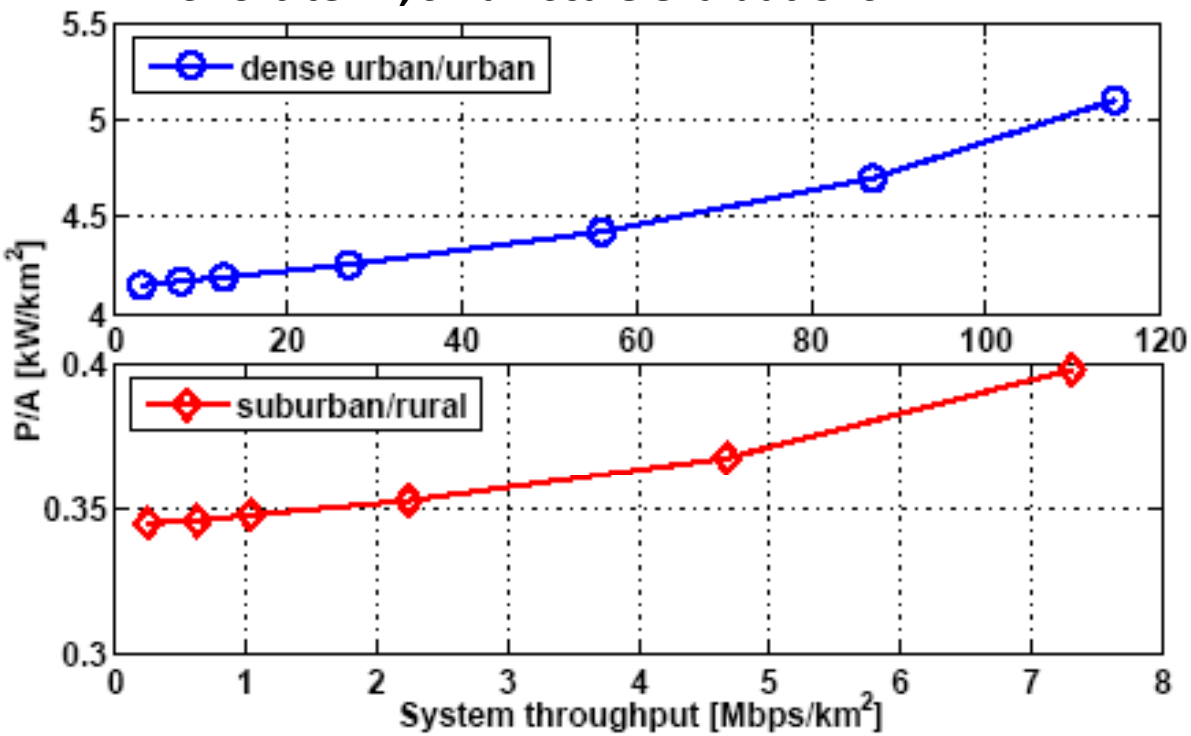
Power Consumption per Unit Area (P/A)

Short-term, small-scale evaluations



• Only a modest dependency of the system throughput on the network power consumption

➔ Due to macro BS power model, which exhibits **poor efficiency at low loads**

- **Aggregation to global scale:**
  - average power per area unit is **0.6 kW/km²**, independent of data usage scenario
  - System operates at **very low loads**: on average **less than 10 %** of resources are utilized, even for the high data usage scenario

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Summary

- The **EARTH E**nergy **E**fficiency **E**valuation **F**ramework **(E^3F)** facilitates a **holistic performance analysis at network level,** comprising:
  - Realistic **base station power model** for various BS types
  - **Long-term traffic** model & **large-scale deployment** model that facilitate **aggregation to global scale**
  - **Energy efficiency metrics** that complement existing performance metrics

- **Energy consumption** of a LTE Rel-8 cellular network **evaluated,** fundamental **energy saving challenges** identified:

  ➔ Vast potential at **low** traffic loads

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